

Technology Stack

Date	03 October 2026
Team ID	FITBUDDY-TEAM
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Jinja2 templates	Provides a lightweight web interface with server-rendered dynamic content.
2	Backend / Server-Side	Python, FastAPI, Uvicorn, Pydantic	Handles routing, validation, API integration and application logic.
3	AI Layer	Google Gemini API, google-generativeai	Generates personalized workout plans, nutrition/recovery tips and feedback-based updates.
4	Database / Data Storage	SQLite with SQLAlchemy	Provides simple persistent storage for user information and plans.
5	Security / Authentication	bcrypt-based password handling and application authentication	Protects registered-user access and stored credentials.
6	Development Tools	Git, VS Code, Python virtual environment	Supports collaborative development and reproducible local setup.